

Bitcoin Exit Dominance in Monetary Coordination Games

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Abstract

Abstract. In a multipolar world with no trusted monetary coordinator, how do rational capital allocators preserve value when sovereign settlement systems are subject to debasement and political capture? The question is not hypothetical: the 2022 freezing of Russian foreign-exchange reserves demonstrated that sovereign settlement is capturable for any holder, central banks included. This paper models the choice as a non-cooperative game (the “Exit Game”) in which allocators choose between capturable settlement systems (“Stay”) and neutral settlement (“Exit”). The model rests on four assumptions: persistent multipolarity, rational self-interest, computational hardness, and the persistence of network effects. Three results follow. First, the payoff advantage of Exit over Stay grows as adoption rises (Proposition 1). Second, no Stay coalition is internally stable once any allocator’s threshold is reached (Proposition 2). Third, the high-Exit region $\{p > p_c\}$ is coalition-proof against returning Stay coalitions, and persistent debasement drives p_c monotonically downward over time (Theorem 1). Together these establish Bitcoin as the primary candidate for neutral settlement derived purely from ongoing incentives (Theorem 2). The formal Exit Game results show viable neutral settlement creates a new form of competition for the existing monetary order.

Revision Note, May 2026. Prepared for journal submission; references updated and tightened. Substantive changes from the February 2026 preprint (SSRN 6299081):

- Theorem 1 is recast in ratchet form. The preprint claimed full absorption at $p = 1$; the revision claims that $\{p > p_c\}$ is coalition-proof and that p_c shifts monotonically downward as the slowly-varying state (K_t, D_t) accumulates (K_t recognition; D_t debasement; §3.5). Permanent holdouts are admitted by construction.
- Adoption is capital-weighted: $p_t = \sum_{i \in \text{Exit}_t} a_i$. Preference falsification (Kuran²³) provides the psychological reasoning for why actors say Stay but allocate Exit. Aggregate p_t counts the allocation, not the stated posture.
- Volatility apparatus is dropped. σ_B , σ_F , and the risk-aversion weight λ_i are removed; monotonicity conditions reduce from (M1)–(M5) to (M1)–(M4). The allocator-side friction is now money illusion (Fisher²²): nominal stability is read as safety and lumpy real outperformance as risk.
- Definition 2 (Capturable System) replaces the preprint’s Capture Surface. D_t in §3.5 now indexes three post-1945 financial-repression channels: debt overhang (IMF²), the inflation–capital-gains tax wedge (Feldstein²¹), and the rate-cap and reserve-requirement toolkit (Reinhart and Sbrancia³).
- Section 5 (formerly Limitations) is folded into the Conclusion; per-assumption failure conditions are retained at conclusion register, with A2’s clause reformulated to a primitive of agent rationality. D_t in §3.2 is flagged as a latent structural index.
- Coalition Instability (Proposition 2) restated as Nash-stability, with P3 (permissionless access) as the load-bearing institutional clause; Bernheim-Peleg-Whinston coalition-proofness is now reserved for Theorem 1 (Ratchet), where it does formal work.

The four assumptions and the central claim of Bitcoin Exit Dominance remain unchanged; the result is restructured into Theorem 2 for clarity.

1. Introduction

A capital allocator with purchasing power to protect faces a choice between two settlement systems. The first is the prevailing one: sovereign currency, custodied by regulated intermediaries, settled through banking and securities infrastructure. Call this *Stay*. The second is a settlement asset that no government issues, no intermediary custodies, and no political body can debase or freeze. Call this *Exit*. This is the *Exit Game*. The question is not which system any one allocator prefers. It is whether the repeated incentives will push a critical mass of capital from Stay to Exit.

While partial exit via gold has existed for thousands of years, true exit only became possible when Nakamoto¹ mined his first block on 3 January 2009. In the genesis block of Bitcoin he encoded that morning’s front-page headline from *The Times* of London: “Chancellor on brink of second bailout for banks.” Since 2009, hashrate on the Bitcoin network has only ever trended higher, despite sovereign mining bans and several hostile hard-fork events. Adoption by ever larger cohorts of capital (ETFs, pension funds, sovereigns) suggests rational allocators continue to defect to Exit.

The usual measure of debasement risk is debt-to-GDP. Public debt is near 100% of GDP and projected at 115% by the early 2030s (IMF²). At that scale, history says the debt gets paid down by debasing the currency, not by taxes (Reinhart and Sbrancia³). High debt is not itself new: what is new is what governments are now committed to spend through the 2030s with no tax base behind it. The United States is stepping back from its role as world military referee, and NATO members pledged in June 2025 to spend 5% of GDP on defense by 2035. The logic in every arms race is the same: spend more or get left behind (Richardson⁴; Powell⁵). AI is the same race: better AI means better weapons, so winning compute means winning force. AI capex in 2026 runs at a multiple of Apollo’s peak GDP share ($\sim 0.7\%$ in 1966), and China is investing at comparable scale through state direction. Neither side can stop without losing the lead (Armstrong, Bostrom, and Shulman⁶). The Exit Game does not require any specific catalyst to work. The emerging AI-military arms race trends above set the backdrop because they are the largest known debasement catalysts today.

The deeper point in the genesis block was the subtle nod to the *Cantillon effect*: in every fiat system, those nearest to issuance capture the wealth transfer that debasement creates (Cantillon⁷). Will aggregate capital flee to a new system where no central issuer can benefit from the Cantillon effect to debase the many to enrich the few? This paper turns that question into a formal game.

2. Related Work

The closest prior policy treatment is Brunnermeier, James, and Landau⁸, who describe the coordination problem verbally without formal derivation and prescribe central bank digital currency. A CBDC remains sovereign-issued, freezable, and debaseable, fails Definition 1, and is an instance of Stay that Theorem 2 covers.

The existing Bitcoin game-theoretic literature concentrates on the mining subgame, which sustains itself through miner skin in the game: hardware built for one purpose, rewards paid in the coin a successful attack would destroy. Two dissents press against this view. Budish⁹ extends his high-frequency-trading rent-seeking critique to Bitcoin mining. Eyal and Sirer¹⁰ identify a selfish-mining limit under which a concentrated pool earns above its honest share by withholding blocks. Neither attack has been profitably executed on chain: shorting Bitcoin at attack scale exceeds available derivatives liquidity, KYC at every fiat off-ramp makes the attacker’s identity legible before they can position, and a pool revealed as malicious loses the hashrate it depends on.

The view of this paper, in line with Szabo¹¹, is the opposite of Budish: mining energy is the unforgeable costliness sustaining neutral settlement.¹² Three results triangulate the intertwined cryptographic and economic hardness Assumption 3 refers to. Biais, Bisière, Bouvard, and Casamatta¹³ show that following the longest chain is a stable equilibrium in the mining subgame. Garay, Kiayias, and Leonardos¹⁴ prove that under honest majority the protocol delivers common prefix, chain quality, and chain growth: the formal cryptographic guarantee that proof-of-work yields consensus. Chen¹⁵ closes the price-security loop in general equilibrium.

3. Model

3.1 Maintained Assumptions

The analysis rests on four assumptions. All subsequent claims derive from these assumptions; rejecting any assumption invalidates the specific claims that depend on it.

Assumption 1 (Multipolarity). Power is split across power centers where no center can exert unilateral control of the settlement layer.

Assumption 2 (Rational Self-Interest). Actors optimize for self-interest. When defection from cooperative agreements is unpunished or unpunishable, actors defect.

Assumption 3 (Computational Hardness). Certain mathematical problems cannot be solved

in practical time by any known algorithm. Digital scarcity and cryptographic custody are implementable.

Assumption 4 (Network Effect Persistence). Past critical mass, switching costs exceed marginal gains of alternatives. An established network attracts further adopters; its lead compounds (Katz and Shapiro¹⁶).

3.2 Definitions

Definition 1 (Neutral Settlement). Bitcoin is the neutral settlement asset of the Exit-Stay game, satisfying Properties P1–P7. Each P_i closes one *attack channel* a coalition of Exit holders could use to profitably return to Stay. Cryptographic and consensus primitives grounding P1–P3 and P5–P6 trace to Nakamoto¹. P1–P7 are assumed throughout, as is the miner-subgame result of §2.

Table 1. The seven attack channels Bitcoin closes. Each row names the property and the attack it blocks.

Closure Property	Attack Channel
P1: Informational security	Physical confiscation
P2: Protocol security	Protocol rule change
P3: Permissionless access	Transaction censorship
P4: Absolute scarcity	Supply inflation
P5: Cheap finality	Prohibitive cost/delay
P6: Neutrality	Governance takeover
P7: Adaptive resilience	Obsolescence

Gold as Neutral Settlement. Gold has been synonymous with neutral settlement across five millennia. Density, divisibility, recognizability, and aesthetic appeal sustained it as money in the market-selection account of monetary emergence (Menger¹⁷). Wampum, cowrie shells, and salt shared the surface properties but failed P4 once industrial production, oceanic shipping, and refined extraction broke their scarcity. Only gold has held P4 across that span. Arslanalp, Eichengreen, and Simpson-Bell¹⁸ trace the post-2022 central-bank gold buying to financial-sanctions risk on the main reserve issuers, not generic flight-to-safety. Gold has four known structural failures: confiscation at handoff, with Executive Order 6102 as precedent; assay risk; settlement cost scaling with mass; and supply inflation at roughly 1.5% annually.

Definition 2 (Capturable System). A monetary system in which at least one of P1–P7 fails for its underlying settlement asset. Fiat, sovereign bonds, and equities have no exit

valve: equities admit no bearer instrument separable from the share, and fiat and bonds have no underlying outside the issuing system. Gold and Bitcoin offer redemption to a self-custodial bearer asset. Gold terminates in physical gold, which retains the four failures above. Bitcoin ETFs can be redeemed in steps to self-custodied Bitcoin, which satisfies P1–P7.

Definition 3 (Exit). The action of moving capital from a capturable system to a neutral settlement asset. The complement is *Stay*: holding capital in the prevailing capturable system.

3.3 Game

The Exit Game is a non-cooperative game in which each capital allocator chooses one of two actions: hold capital in the prevailing capturable system (Stay) or move it to a neutral settlement asset (Exit). Exit is not free. Early exit to Bitcoin requires a social cost to act against peer consensus. Depending on jurisdiction and employment, exit to Bitcoin can also cost real legal or reputational damage.

Players. N is the population of capital allocators, weighted by their share a_i of total capturable capital. $\kappa_i > 0$ is how much social cost allocator i pays for moving before peers. $\rho_i > 0$ is how exposed allocator i is to state penalty for the move itself. Fiduciaries answering to boards, committees, or constituencies sit at the high end of κ_i . Individuals and unilateral sovereigns sit lower. Both weights vary continuously across allocators.

Objective. Preserve real purchasing power. Preference between capturable and neutral assets follows from Exit Game payoffs alone.

Rules. Each allocator’s per-period action is binary: Exit or Stay. The aggregate p_t evolves continuously because allocators are heterogeneous in their thresholds and cross at different moments. Per-player actions are reversible: a player can re-Stay if p_t retreats below their personal threshold. Switching assets incurs friction; positions are sticky. As p rises, Exit pays more and Stay pays less (§3.4).

Dynamics. Recognition that capturable assets erode in real terms grows; debasement keeps accumulating. As both rise, p_t trends upward (§3.5).

Payoffs. The game is $G = (N, S, u)$ in standard non-cooperative form (Fudenberg and Tirole¹⁹), with per-period action set $S_i = \{\text{Exit}, \text{Stay}\}$, capital weights $\sum_i a_i = 1$, and aggregate adoption $p_t = \sum_{i \in \text{Exit}_t} a_i$. Each player’s strategy is a threshold $p_i^* \in [0, 1] \cup \{+\infty\}$: Exit when $p_t > p_i^*$, Stay when $p_t \leq p_i^*$. The per-player action is binary and reversible at the player level; the aggregate p_t moves continuously across the heterogeneous threshold

distribution and may itself drawdown. The monotone object is the critical mass p_c , which shifts downward as (K_t, D_t) (recognition; debasement) accumulate (§4.3).

$$u_i^E(p) := R_B(p) - \kappa_i K_A(p) - \rho_i R_A(p),$$

$$u_i^S(p) := R_F - K_N(p).$$

- $R_B(p)$: real return on the neutral asset.
- R_F : return on capturable assets.
- $K_A(p)$: conformity cost of acting before peers (Katz and Shapiro¹⁶).
- $R_A(p)$: realization and regulatory friction; gains tax, fees, *etc.*
- $K_N(p)$: non-adoption penalty.

Weights are strictly positive and differ across allocators. R_B , K_A , R_A , K_N are continuous and bounded.

Table 2. Normal-form payoff matrix for player i , with u_i^E , u_i^S as defined above.

	Others: Stay	Others: Exit
Player i : Stay	$u_i^S(0)$	$u_i^S(p)$
Player i : Exit	$u_i^E(0)$	$u_i^E(p)$

3.4 Monotonicity Conditions

Milgrom and Shannon²⁰ showed that when the payoff gap between two options widens as some parameter rises, the player's best response moves toward the favored option and does not return. This paper applies that result with adoption p as the parameter and Exit as the favored option: as more actors exit, Exit's gap over Stay widens. Four conditions on the payoff components secure that widening.

- (M1) $R'_B(p) > 0$: Bitcoin's return rises with adoption (network effects, Assumption 4)
- (M2) $K'_A(p) < 0$: The cost of switching to Bitcoin falls as more allocators switch
- (M3) $R'_A(p) < 0$: Friction eases as adoption normalizes
- (M4) $K'_N(p) > 0$: The cost of staying in capturable assets rises as competitors exit

These conditions hold on average over long horizons rather than at every instant.

3.5 Threshold Drift

Sudden seizure (capital controls, asset freezes, hyperinflation) is the visible failure mode of the capturable system. The actual failure mode is slower: continuous extraction of real

wealth. Debasement raises the price level. Equities, real estate, and similar assets inflate nominally as they absorb a share of the newly created money. Holders are then taxed on the nominal gain. After tax, real returns are negative (Feldstein²¹). Public debt keeps growing; private claims bear the hidden inflation tax to inflate away the real debt burden. Multipolarity (Assumption 1) leaves debasement unchecked, so $R_F < 0$ in real terms persists. Allocators under money illusion (Fisher²²) read nominal stability as safety and lumpy real outperformance as risk, so the erosion goes uncorrected.

The macro state evolves on two slowly-varying indices:

- $K_t \in [0, 1]$ (*recognition*): the share of capital allocators who recognize capturable assets as financially repressed. Recognition accumulates gradually rather than arriving with a discrete capture event.
- $D_t \in [0, \infty)$ (*debasement*): a latent debasement-intensity index summarizing three post-1945 financial-repression channels: debt-to-GDP overhang (IMF²), the inflation–capital-gains tax wedge (Feldstein²¹), and the rate-cap and reserve-requirement toolkit (Reinhart and Sbrancia³).

Both K_t and D_t are monotonic over the structural horizon. Short-horizon reversals are admitted as fluctuations around the trend: generational forgetting in K_t , hawkish tightening episodes in D_t .

Let $\Delta_i(p; K, D) := u_i^E(p) - u_i^S(p)$ denote player i 's Exit minus Stay payoff gap at adoption p and macro state (K, D) . Each allocator's threshold depends on (K, D) :

$$p_i^*(K, D) := \inf\{p \in [0, 1] : \Delta_i(p; K, D) > 0\}.$$

Assume the sign conditions

$$\frac{\partial p_i^*}{\partial K} \leq 0, \quad \frac{\partial p_i^*}{\partial D} \leq 0.$$

These hold strictly for at least some capital allocators. Higher D_t lowers R_F in real terms. Higher K_t erodes $\kappa_i K_A$ and $\rho_i R_A$. Both effects raise $\Delta_i(0; K, D)$ and push p_i^* down.

The critical mass invoked in Assumption 4 is the smallest adoption level at which at least one allocator's threshold has been crossed:

$$p_c(K, D) := \inf\left\{p \in [0, 1] : \sum_{i: \Delta_i(p; K, D) > 0} a_i > 0\right\}.$$

Above p_c , the M1–M4 feedbacks (Katz and Shapiro¹⁶) prevent Stay coalitions from reforming (§4.2). The endpoint is dominance in the network-effects sense, not full adoption $p = 1$. For each (K, D) , $\{p > p_c\}$ is coalition-proof against return-coalitions. As (K_t, D_t) accumulate, p_c shifts downward by monotone comparative statics (Milgrom and Shannon²⁰): $p > p_c$

holds even when p_t is flat, and the coalition-proof region expands (§4.3).

The model's claims are structural and directional: existence of thresholds $\{p_i^*\}$, sign of comparative statics, and direction of shift in p_c under accumulating (K_t, D_t) . It does not predict specific values, tipping times, or saturation levels.

An adoption level p is an equilibrium of the Exit Game when no allocator, alone or in any group that holds together, gains by switching.

4. Results

4.1 Monotone Exit Advantage

First, each actor's payoff gap from Stay to Exit widens as more actors exit. Every actor therefore has a threshold above which Exit becomes the only rational move. Hold the slowly varying state (K, D) fixed throughout this subsection. By convention, $p_i^* = +\infty$ when Δ_i stays non-positive on $[0, 1]$.

Proposition 1 (Monotone Exit Advantage). *Under Assumptions 1–4 and maintained conditions (M1)–(M4):*

- (i) *The payoff differential $\Delta_i(p)$ is strictly increasing in p for all player types.*
- (ii) *For each player i there exists a threshold $p_i^* \in [0, 1] \cup \{+\infty\}$ such that Exit is the unique best response for $p > p_i^*$.*

Proof. Define the payoff differential and take its derivative in p :

$$\begin{aligned}\Delta_i(p) &= u_i(\text{Exit}) - u_i(\text{Stay}) \\ &= [R_B(p) - R_F] - \kappa_i K_A(p) - \rho_i R_A(p) + K_N(p), \\ \frac{d\Delta_i}{dp} &= R'_B(p) - \kappa_i K'_A(p) - \rho_i R'_A(p) + K'_N(p).\end{aligned}$$

Under (M1)–(M4), each term is signed: $R'_B(p) > 0$ and $K'_N(p) > 0$ strictly, while $-\kappa_i K'_A(p) \geq 0$ and $-\rho_i R'_A(p) \geq 0$ for $\kappa_i, \rho_i > 0$. Therefore $d\Delta_i/dp > 0$: the advantage of Exit rises strictly with p . Each player's threshold $p_i^* := \inf\{p \in [0, 1] : \Delta_i(p) > 0\}$ specializes the §3.5 definition with (K, D) fixed. For actors with $p_i^* \in [0, 1]$, the intermediate value theorem applied to the continuous Δ_i delivers $\Delta_i(p_i^*) = 0$, and for $p > p_i^*$ Exit strictly dominates.

4.2 Coalition Instability

§4.1 gave each actor a threshold. The group implication is sharper: no Stay coalition is self-enforcing once any one member has crossed. The first defector profits unilaterally; the actors left behind face a higher exit advantage than before.

Proposition 2 (Coalition Instability). *Under Assumption 2 and Property P3, no Stay coalition C containing at least one actor with a crossed threshold ($p > p_j^*$ for some $j \in C$) is Nash-stable.*

Proof. Consider a coalition $C \subseteq N$ maintaining Stay. For C to be Nash-stable, no member $j \in C$ can profit from unilateral defection. By Proposition 1, $u_j(\text{Exit}) > u_j(\text{Stay})$ when p exceeds p_j^* . Property P3 (permissionless access) closes the institutional defection channel: no counterparty can refuse the trade and no clearing intermediary can foreclose settlement, so the only remaining friction is the regulatory cost already baked into each actor's threshold via R_A . Under Assumption 2, the crossed-threshold member defects: for at least one $j \in C$, $u_j(\text{defect from } C) > u_j(\text{remain in } C)$. The coalition therefore fails Nash.

Proposition 2 covers coalitions with at least one crossed-threshold member. Permanent holdouts ($p_j^* = +\infty$), admitted by Proposition 1, lie outside its scope: no private reason to defect. The aggregate share of capital held for ideological rather than rational reasons (Assumption 2) is not fixed: preferences adjust as new information arrives. The real-world case is regulatory coercion holding the coalition together: capital controls, fiduciary mandates, jurisdictional penalties. P1 limits what coercion can do. An asset controlled by secret information, not physical possession, can be seized only by forcing each holder to disclose it one at a time. Cost grows with the number of holders, not the size of any one holding. Preference falsification (Kuran²³) then separates what holders say from what they do. Aggregate p_t tracks what they do. Once enough private exits register, public Stay breaks.

4.3 Ratchet

§4.1 gave each actor a threshold; §4.2 showed coalitions cannot hold once any one member crosses. §4.3 lets the slowly varying state (K_t, D_t) move. Per-period actions stay binary and reversible at the player level. What changes over time is the macro state, and with it the threshold profile $\{p_i^*(K_t, D_t)\}$ and the critical mass $p_c(K_t, D_t)$. Recognition K_t ratchets rather than oscillates: once private allocations reveal previously concealed Exit preferences, the social cost structure sustaining prior public Stay no longer holds (Kuran²³). Debasement D_t does not reverse on the structural horizon (Reinhart and Sbrancia³). Both push p_c downward. The result is not that any specific player is locked in; it is that the floor a

returning Stay coalition would have to clear keeps dropping. Two claims follow. First, above p_c no returning Stay coalition can form: a candidate with a crossed-threshold member is broken by Proposition 2; a candidate without one contains only allocators not moving to Exit. Second, p_c is non-increasing in t .

Theorem 1 (Ratchet). *Under Assumptions 1–4, maintained conditions (M1)–(M4), and the §3.5 sign conditions: (a) $\{p > p_c\}$ is coalition-proof^{24,25} against returning Stay coalitions. (b) Under monotonic accumulation of (K_t, D_t) , the critical mass $p_c(K_t, D_t)$ shifts monotonically downward in t .*

Proof. (a) *Coalition-proofness of $\{p > p_c\}$.* If $p > p_c(K, D)$, the crossed-threshold set

$$H(p) := \{i : \Delta_i(p; K, D) > 0\}$$

has positive capital weight. Any perturbation attempting to coordinate a return to high-Stay above p_c is a candidate Stay coalition C and falls into one of two cases. If $C \cap H(p)$ is nonempty, Proposition 2 applies: the crossed-threshold member defects under P3, so C fails Nash and therefore fails BPW self-enforcement. If $C \cap H(p)$ is empty, C consists entirely of uncrossed-threshold capital. These actors already prefer Stay at the current p and do not move toward Exit; no candidate Stay coalition expands above p_c . The system is coalition-proof above p_c . As (K_t, D_t) accumulate, the finite-threshold Stay pool shrinks and the coalition-proof region expands. Drawdowns and cascade pauses along p_t are admitted; price drawdowns are not equivalent to drawdowns in p_t , since allocators below target weight can rebalance during decline. A permanent stall would require two conditions to hold jointly: p_t plateauing below p_c indefinitely, and p_c failing to fall. The second condition is ruled out by part (b), so any p_t plateau is finite-duration.

(b) *Drift of p_c .* Because p_c is defined from the individual thresholds, the §3.5 sign conditions transfer directly to p_c :

$$\frac{\partial p_i^*}{\partial K} \leq 0, \quad \frac{\partial p_i^*}{\partial D} \leq 0, \quad \frac{\partial p_c}{\partial K} \leq 0, \quad \frac{\partial p_c}{\partial D} \leq 0.$$

Under monotonic accumulation of (K_t, D_t) , $p_c(K_t, D_t)$ is non-increasing in t .

4.4 Bitcoin Exit Dominance

The three preceding results all hold simultaneously in the Exit Game under the same hypotheses.

Theorem 2 (Bitcoin Exit Dominance). *Under the four maintained assumptions A1–A4 of §3.1 (multipolarity, rational self-interest, computational hardness, network-effect persistence), the seven settlement properties P1–P7 of §3.2, the Exit Game G of §3.3, the monotonicity conditions (M1)–(M4) of §3.4 governing payoff derivatives in p , and the §3.5 sign conditions, the capital-weighted adoption process satisfies all three of:*

- (i) *the payoff advantage of Exit over Stay is strictly increasing in p for every player type (Proposition 1);*
- (ii) *no Stay coalition with at least one crossed-threshold member is self-enforcing (Proposition 2);*
- (iii) *$\{p > p_c\}$ is coalition-proof, and p_c shifts monotonically downward under accumulating (K_t, D_t) (Theorem 1).*

Proof. Proposition 1 gives (i). Proposition 2, with (i) supplying the per-defector incentive, gives (ii). Theorem 1 gives (iii). The conjunction of (i)–(iii) is the Bitcoin Exit Dominance claim.

5. Conclusion

The Exit Game treats capital allocators, not sovereign issuers, as the agents who decide what counts as settlement. The long-run outcome depends on whether an asset exists that no government issues. Once such an asset exists and satisfies P1–P7, the question is no longer whether allocators can leave the capturable system but whether continued debasement pushes adoption past p_c .

All capital allocators choose reserve assets on perceived risk and ability to store value. Since the Global Financial Crisis central banks have been accumulating gold. Smaller allocators, with higher risk tolerance and lower position size, can shift first into an asset that holds gold’s properties without its four structural failures. §3.5 predicts that continued debasement keeps moving more allocators out of fiat.

A1–A4 are each independently load-bearing. A1 (persistent multipolarity) fails only if rival regimes coordinate and preference falsification no longer holds; rival seignorage is what holds the multipolar floor. A2 (rational self-interest) fails only if actors systematically forgo positive expected returns from unpunished defection. A3 (computational hardness) fails if either the underlying cryptography breaks and the protocol cannot replace it, or the economic incentive sustaining the mining subgame falls below the level §2 requires. A4 (persistence of network effects) fails only if the incumbency account of network goods is disproved.

The Bitcoin Exit Dominance claim is new and depends on two conditions outside this paper’s scope. First, no other consensus ledger holds all of P1–P7: alternatives must either

fail one of the properties or fail to draw adoption. Second, autonomous AI agents that need a settlement asset choose Bitcoin rather than invent a new settlement asset. A companion paper in preparation addresses the first across consensus ledgers; a second extends neutral settlement to autonomous agents.

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Conflict of Interest. The author holds bitcoin and operates bitcoingametheory.com.

Table A1. Notation used throughout.

Symbol	Definition
G	Settlement game (N, S, u)
N	Set of capital allocators
S_i	Strategy set: {Exit, Stay}
u_i	Utility function
a_i	Actor i 's share of total capturable capital, $\sum_i a_i = 1$
p_t	Aggregate capital share at Exit at time t : $\sum_{i \in \text{Exit}_t} a_i$
$p_i^*(K, D)$	Threshold for player i at slowly varying state (K, D)
$p_c(K, D)$	Critical-mass threshold at (K, D)
K_t	Financial-repression recognition share, $K_t \in [0, 1]$
D_t	Debasement-intensity index, $D_t \in [0, \infty)$
$\Delta_i(p; K, D)$	Payoff differential: $u_i(\text{Exit}) - u_i(\text{Stay})$ at (K, D)
$R_B(p)$	Expected real return on neutral settlement asset
R_F	Expected real return on capturable assets
$K_A(p)$	Conformity cost of acting before peers
$R_A(p)$	Realization and regulatory friction (decreases as adoption normalizes)
$K_N(p)$	Non-adoption penalty
κ_i, ρ_i	Conformity weight, friction weight (all > 0 strictly)
(M1)–(M4)	Monotonicity conditions (§3.4)

6. Notes and References

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12. *Note on Budish (§2)*. The mining subgame imposes real costs that are a legitimate object of inefficiency analysis. The viewpoint of this paper is that those costs are not inefficiency in the rent-seeking sense but unforgeable costliness (Szabo¹¹) sustaining neutral

settlement: the specific property Bitcoin holders are paying to obtain. A full reconciliation with Budish’s bound is beyond the scope of one paper and is left to subsequent work.

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25. *Note on coalition-proofness vocabulary (§4.3).* The claim is coalition-proofness in the Bernheim-Peleg-Whinston sense, not stochastic stability in the Young or Kandori-Mailath-Rob sense; the paper does not specify a perturbed best-response dynamic.